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THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

Screening Test - Bhaskara Contest

(NMTC- at **JUNIOR LEVEL** IX & X Standards)

Saturday, 25th August 2007.

Note:

1) *Fill in the answer sheet with your Name, Class, and Section etc. in the specified places.*
2) *Only one of the choices A,B,C,D is correct for each question. Write the alphabet of your choice in the answer sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).*
3) *For each correct response you get 1 mark; **for each incorrect response you lose ¼ mark.***
4) *Diagrams are only visual aids; they are not drawn to scale.*
5) *You are free to do rough work on separate sheets.*
6) *Duration of the test: 2 p.m. to 4. p.m.- 2 hours*

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1. In a polygon there are 6 right angles and the remaining angles are all equal to 200^0 each. The number of sides of the polygene is

   (A) 15 **(B) 12** **(C) 9** **(D) 23**

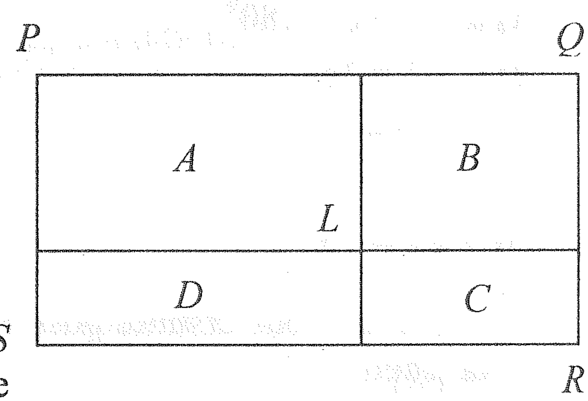
2. Two natural numbers differ by 41. The bigger number is greater than 30 times the smaller number plus 10 The smaller number is

   (A) 11 **(B) 2007** **(C) 71** **(D) 1**

3. The value of $\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\left(1-\frac{1}{4^2}\right)\ldots\ldots\ldots\ldots\left(1-\frac{1}{2007^2}\right)$ is

   (A) $\frac{2008}{2007}$ **(B)** $\frac{1004}{2007}$ **(C)** $\frac{2007}{2008}$ **(D) 1**

4. *PQRS* is a rectangle of area 2000 cm^2. Two lines parallel to the sides are drawn to cut the rectangle into four rectangles of areas A, B, C, D, Given A = 1000 cm^2, B = 500 cm^2, C = 300 cm^2, D = 200 cm^2 which of the following is possible?



   (A) Such a division is not possible
   (B) Exactly two such divisions are possible
   (C) At least two such divisions are possible
   (D) In such a division the point L lies on one of the diagonals

5. If (x,y) is a solution set of the system of equations $xy = 8$ and $x^2y + y^2x + x + y = 54$, then $x^2 + y^2 =$

   (A) 62 **(B) 46** **(C) 20** **(D) 100**

6. The number of natural numbers 'n' for which $\frac{15n^2+8n+6}{\ldots}$ is a natural is

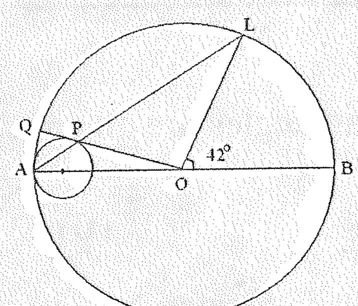
7. If a, b, c are real ; $a \neq o$, $b \neq o$, $c \neq o$ and $a+b+c \neq o$ and $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=\frac{1}{a+b+c}$

then $(a+b)(b+c)(c+a)=$

(A) 1 **(B) $3abc$** **(C) 0** **(D) abc**

8. In a rectangle the length is x units more than its breadth .If its length is increased by y units and its breadth is decreased by z units, the area is unaltered .The breadth of the rectangle is

(A) $\frac{(x+y)z}{y-z}$ **(B)** $\frac{z}{x+y}$ **(C)** $\frac{(x-y)z}{y+z}$ **(D)** $\left(\frac{x+y}{x-y}\right)Z$

9. 

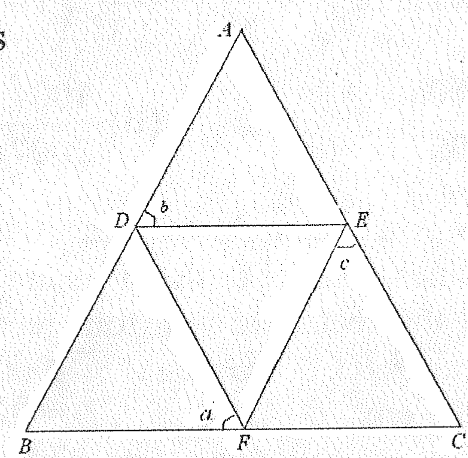
Two circles touch internally at A. AOB is the diameter of the bigger circle of centre O. OPQ is a tangent to the smaller circle. APL is a straight line. If $\angle BOL = 42^0$. then $\angle LOQ =$

(A) 84^0 **(B) 104^0** **(C) 90^0** **(D) 117^0**

10. The units digit of $3^{2007} \times 7^{2008} \times 13^{2009}$ is

(A) 3 **(B) 1** **(C) 9** **(D) 7**

11. In the adjoining figure, AB = AC. DEF is an equilateral triangle. Then



(A) $a+b+c=180^0$

(B) $a+b=2c$

(C) $a=\frac{b+c}{2}$

(D) $a+c=2b$

12. a, b, c, d are four distinct positive real numbers such that $a \gtrsim b$, $c < d$, $b > c$ and $d < a$. Then

(A) $b < d < a$ always **(B) $d < b < a$ always**

(C) $d > c$ and $d < b$ **(D) a is the greatest always**

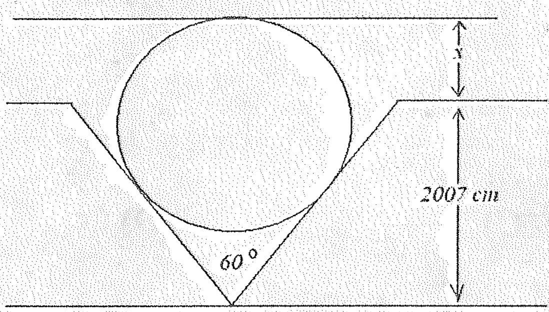
13. $x^{n+1} - x^n - x + 1$ is exactly divisible by $(x-1)^2$ if n is

(A) an odd positive integer **(B) an even positive integer**

(C) an odd prime **(D) any positive integer**

14. If a, b, c, d, are non zero positive reals and $a + b + c + d = s$, then the minimum value of $\left(\frac{s}{a}-1\right)\left(\frac{s}{b}-1\right)\left(\frac{s}{c}-1\right)\left(\frac{s}{d}-1\right)$ is

(A) 4 **(B)** S **(C)** 81 **(D)** S^4

15. 

In the adjoining figure the diameter of the circle is 2007 cm. Then $x =$

(A) $\frac{2007}{2} cm$ **(B)** $\frac{2007}{4} cm$

(C) $2007 cm$ **(D)** $\frac{3}{2} \times 2007 cm$

16. The number of real solutions of the equation $1 + x + x^2 + x^3 = x^4 + x^5$ is

(A) 1 **(B)** 5 **(C)** 4 **(D)** 3

17. A two digit number is increased by 20% when its digits are reversed. Then the sum of the digits of the number is

(A) 2 **(B)** 7 **(C)** 9 **(D)** 8

18. If a, b, c, d are positive integers such that $a = bcd$, $b = cda$, $c = abd$ and $d = abc$, the value of $\frac{(a+b+c+d)^4}{(ab+bc+cd+da)^2}$ is

(A) 16 **(B)** 1 **(C)** 3^4 **(D)** $\frac{4^4}{3^2}$

19. a, b, c are three numbers satisfying the following conditions.
(i) $abc \neq 0$ (ii) $a + b + c = abc$ (iii) $(a + b)(b + c)(c + a) \neq 0$ and
(iv) $\frac{a+b}{1-ab} + \frac{b+c}{1-bc} + \frac{c+a}{1-ca} = kabc$, then $k =$

(A) 0 **(B)** 1 **(C)** -1 **(D)** 2

20. The sum of the fourth powers of the roots of the equation $x^3 - x^2 - 2x + 2 = 0$ is

(A) 1 **(B)** 5 **(C)** 9 **(D)** 13

21. If $\tan\theta$ and $\tan\phi$ are the roots of the quadratic equation $x^2 - Ax + B = 0$ and $\cot\theta$ and $\cot\phi$ are the roots of $x^2 - Cx + D = 0$ then $CD =$

(A) AB **(B)** $\frac{1}{A-B}$ **(C)** $\frac{A}{B^2}$ **(D)** $\frac{A}{B}$

22. A, B, C can walk at the rates 3, 4, and 5km an hour respectively. They start from a place at 1,2,3 hrs. respectively when B catches A. B sends him back with a message

23. The units digit of $(1 + 9 + 9^2 + 9^3 + \ldots\ldots\ldots + 9^{2007})$ is

(A) 1 (B) 9 (C) 0 (D) 8

24. The sides of a triangle are integers. The perimeter is 8. the area is

(A) $\sqrt{8}$ (B) 8 (C) 5 (D) 12

25. If $x + y + z = 2007$, $xy + yz + zx = 4011$ then the value of $\frac{1}{1-x} + \frac{1}{1-y} + \frac{1}{1-z}$ is

(A) 1 (B) 2008 (C) $\frac{1}{2008}$ (D) 0

26. A boy multiplies 423 by a certain number and obtains 65589 as his answer. If both the fives are wrong but the other digits are correct, the right answer is

(A) 63489 (B) 60489 (C) 62689 (D) 60689

27. Let A denote the ratio of the volumes of a cube to that of the sphere which will fit inside the cube. Let B be the ratio of the volume of a cube inscribed in a sphere to the that of the that of the sphere, then

(A) $A^2 = 27\,B^2$ (B) $B^2 = 27\,A^2$ (C) $A^2 = 9B^2$ (D) $B^2 = 9A^2$

28. If the average of 'n' observations (where n is odd) arranged in ascending order is a, the average of the first $\left(\frac{n+1}{2}\right)$ observation is b and that of the last $\left(\frac{n+1}{2}\right)$ observations is c, then the $\left(\frac{n+1}{2}\right)$ th observation is

(A) $(n+1)(b+c) - na$ (B) $\frac{n(a+b+c)}{3}$ (C) $\frac{(n+1)(b+c)-2na}{2}$ (D) $\frac{a+b+c}{3n}$

29. The ratio between a two digit number and the sum of the digits of that number is $a{:}b$. If the digit in the units place is n more than the digit in the tens place then the number in the tens' place is

(A) $\frac{n(a-b)}{11b-2a}$ (B) $\frac{n(a+b)}{11b-2a}$ (C) $\frac{2n(a-b)}{11b-2a}$ (D) $\frac{a-b}{n(11b-2a)}$

30. For the equation $\frac{(x-a)(x-b)}{x-a-b} = \frac{(x-c)(x-d)}{x-c-d}$

(A) $x = \frac{ab(c+d)-cd(a+b)}{ab-cd}$ is the only solution

(B) $x = \frac{ab(c+d)-cd(a+b)}{ab-cd}$ is one of the solutions